

Surya Duraivenkatesh

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EDUCATION

University of Washington

Sep. 2025 – Jun. 2028

B.S. Computer Science, B.S. Mathematics | Cumulative GPA: 3.88/4.00 | CSE GPA: 3.92/4.00

Seattle, WA

- Purple & Gold Scholarship; Out-of-state Allen School Direct Admit; YC Startup School 2026
- **Relevant Coursework:** CSE 421: Algorithms, CSE 447: Natural Language Processing, CSE 493: Deep Learning, CSE 589: Software Entrepreneurship (Graduate), MATH 334: Accelerated Honors Advanced Calculus, CSE 332: Data Structures & Parallelism, CSE 331: Software Design & Implementation

EXPERIENCE

Undergraduate Research Assistant

Mar. 2026 – Present

RAIVN Lab, University of Washington – Mentors: Ethan Shen, Dr. Ali Farhadi

Seattle, WA

- Investigated incorrect-trace supervision for LLM fine-tuning; found **training–evaluation relevance can outweigh trajectory correctness** and incorrect traces generalize comparably on unseen AIME problems
- Designed selective-SFT partial supervision; **25% masking reached 73.33% MATH-500** vs. **72.00%** correct-only SFT, with up to **70%** masking remaining competitive
- Ran **50+** full-parameter SFT runs + a **24-point** LoRA sweep on **0.6B–32B** Qwen; Qwen3-32B → Qwen3-8B distillation: **200K** SFT traces + **6K** rollouts; built reproducible multi-GPU train/eval infra over **4K+ H200 GPU-hours**

Campus Lead, Student Collective

Aug. 2026 – Present

OpenAI

Seattle, WA

- Selected for OpenAI’s inaugural **140-student** U.S./Canada cohort; lead UW workshops, studio hours, and showcases to expand AI access across disciplines

Software Engineering Intern

Aug. 2023 – Mar. 2026

Dasion Corporation (NSF-backed)

- Built OpenAI API automation for personalized outreach, grant discovery, NSF proposal drafting, and lead targeting, increasing reply rates **3–4×**
- Built campaign app for outreach scheduling, lead management, campaign-data import, and performance tracking

Logistics Team Member

Jan. 2026 – Present

DubHacks

Seattle, WA

- Launched DubHacks’ 2026 Guest Speaker Initiative; coordinate logistics/sponsorships for the Pacific Northwest’s largest collegiate hackathon (**1,000+** attendees) and led executive outreach to Uber, GoDaddy, Zapier, Atlassian

President

2021 – 2025

XR EDU Club

San Francisco Bay Area

- Led HackThePlanet 2024–2025 for **70+** participants across 8 schools and **\$5K+** in prizes; secured Adobe, PG&E, Art of Problem Solving, and Wolfram support
- Samsung Solve for Tomorrow State Finalist (2024–25); Games for Change XR Innovation National Winner (2022)

SELECTED PROJECTS

Y86-64 Assembler and Simulator | C, Assembly

Aug. 2024 – Jun. 2025

- Built Y86-64 assembler and instruction-level C simulator with instruction execution, register/memory tracking, and processor-state inspection

Retinoblastoma Detection via Interpretable AI | Python, TensorFlow, NumPy

Aug. 2022 – Jun. 2023

- Benchmarked **5** transfer-learning architectures for retinoblastoma detection, achieving **0.989** Dice / **0.979** IoU with VGG16; applied SHAP/LIME; presented at IEEE CAI 2023

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Bash, Assembly

Systems / Distributed: Linux, Git, Slurm, Ray, Apptainer, CUDA, vLLM, SGLang, Tensor/Data/Context Parallelism (TP/DP/CP), ZeRO, FlashAttention

ML / LLM: PyTorch, Hugging Face Transformers, Axolotl, slime, Megatron-LM, LoRA, SFT, OPD, GRPO

Data / Tools: NumPy, TensorFlow, Pandas, Matplotlib, OpenCV